



Women suicide terrorism: The strategic and tactical logic of civilian killings

Seung-Whan Choi

University of Illinois at Chicago

ABSTRACT

A growing body of research indicates that women are increasingly involved in suicide attacks. These studies suggest that women, due to their seemingly innocent appearance, can more effectively target civilians rather than soldiers. However, these studies often overlook the strategic and tactical motives behind these attacks. This study argues that gender has become an important factor as terrorist organizations adopt new strategies—specifically, targeting civilians—and new tactics, such as using women to inflict high civilian casualties. Consequently, women are less likely to target well-defended security personnel, like soldiers and police, and are more likely to attack civilian targets compared to men. This gender difference in targeting behavior is supported by a statistical analysis of 6,127 suicide bombing incidents from 1980 to 2016. Therefore, the counterterrorism community needs to develop effective measures to address the new security challenges posed by female suicide bombers.

KEYWORDS

Civilian killings; hard/ security targets; soft/ civilian targets; strategic and tactical logic; terrorism; women suicide bombers

RESUMEN

Existe un número creciente de investigaciones que indican que las mujeres están cada vez más implicadas en ataques suicidas. Estos estudios sugieren que las mujeres, debido a su aspecto aparentemente más inocente, pueden atacar con mayor eficacia a civiles que a soldados. Sin embargo, estos estudios, con frecuencia, pasan por alto los motivos estratégicos y tácticos que se encuentran detrás de estos ataques. Nuestro estudio sostiene que el género se ha convertido en un factor importante a medida que las organizaciones terroristas adoptan nuevas estrategias (específicamente, atacar a civiles) y nuevas tácticas, tales como usar mujeres con el fin de causar un alto número de bajas civiles. En consecuencia, las mujeres tienen menos probabilidades de atacar a personal de seguridad, que cuenta con mayores posibilidades en materia de defensa, tales como soldados y policías. Por el contrario, es más probable que puedan atacar objetivos civiles en comparación con los hombres. Esta diferencia de género en el comportamiento de los atacantes está respaldada por un análisis estadístico llevado a cabo sobre 6127 incidentes de atentados suicidas con bombas entre 1980 y 2016. Por lo tanto, la comunidad antiterrorista necesita desarrollar medidas efectivas para abordar los nuevos

CONTACT Seung-Whan Choi ✉ whanchoi@uic.edu 📧 Professor, Department of Political Science (MC 276), University of Illinois at Chicago, 1007 W Harrison St, Chicago, IL 60607-7137

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desafíos en materia de seguridad que plantean las mujeres bomba suicidas.

RÉSUMÉ

Selon un pan croissant de la recherche, les femmes prennent de plus en plus part à des attentats-suicides. Ces études suggèrent que les femmes, que l'on imagine inoffensives, réussissent mieux à cibler des civils que des soldats. Cependant, ces études négligent souvent les motivations stratégiques et tactiques derrière ces attaques. Cette étude affirme que le genre devient un facteur important, car les organisations terroristes adoptent de nouvelles stratégies (prendre des civils pour cibles) et de nouvelles tactiques, comme le recours aux femmes pour tuer beaucoup de civils à la fois. Par conséquent, par rapport aux hommes, les femmes ont moins de chances de cibler les forces de sécurité bien protégées, comme les soldats ou la police, mais davantage de s'attaquer à des cibles civiles. Cette différence genrée dans le comportement de ciblage se trouve confirmée par une analyse statistique de 6 127 attentats-suicides à la bombe entre 1980 et 2016. Aussi la communauté de lutte contre le terrorisme doit-elle mettre au point des mesures efficaces pour répondre aux nouveaux défis de sécurité posés par les femmes kamikazes.

What role do women play in suicide terrorism?¹ The attacks in Paris and Brussels killed around 130 people in the French capital in 2015 and approximately 30 people at an airport and a metro station in the Belgian capital in 2016. In these suicide attacks, women played a nontrivial role by providing logistic support, hideouts, and preparations for follow-up attacks (Kiss 2016). In other instances, women transformed themselves into lethal weapons, carrying out bombings against security officers and civilians. For example, in July 2017, a woman detonated explosives amidst a group of Iraqi troops who were flushing out the Islamic State in Iraq and Syria (ISIS) fighters from the city of Mosul. To avoid arousing suspicion, she held an infant in her arms. There were over 20 instances of women carrying out suicide attacks in the city over the past two weeks (Roberts 2017). Similar incidents involve women targeting civilian locations such as schools, shopping centers, and funerals. For instance, three women affiliated with Boko Haram set off explosives at a market and a refugee camp in northeastern Nigeria in 2017, resulting in the deaths of 28 people and wounding 82 others (Guardian 2017). These examples illustrate the critical role that women have played in suicide terrorism over the past decade.

Intrigued by a series of women suicide attacks occurring across the world, scholars have discussed whether suicide bombers, in general, and women bombers, in particular, tend to strike security targets more often than civilian ones. A group of scholars believe that suicide bombers prefer

to target well-guarded security sites, such as military checkpoints and barracks, rather than relatively unprotected civilian sites in public areas (e.g., Berman and Laitin 2008; Crenshaw 2007; Hemmingby 2017).² Piazza (2020) emphasizes that “suicide attacks are both more frequently deployed against hard targets, and are more often successfully executed against hard targets. As previous literature might suggest, this finding is not a product of foreign military interventions.”

Another group of scholars contend that suicide terrorism targets civilians rather than security targets (e.g., De la Calle 2017; Shapiro 2013; Zenn and Pearson 2014). These scholars often conduct case-oriented research, examining instances of suicide attacks in countries such as Iraq (Hicks et al. 2011; Seifert and McCauley 2014), Israel and Palestine (Harrison 2006), and Pakistan (Bhatti et al. 2011). A few scholars use statistical analysis to gather evidence for suicide bombers attacking civilian targets (e.g., Asal, Brown, and Schulzke 2015; Asal, Phillips, and Rethemeyer 2022; Bhatti et al. 2011; De la Calle 2017; Hsiao-Rei Hicks et al. 2011; Pearson 2018; Welsh 2024).³

Several scholars have provided new insights into the debate by incorporating gender into discussions about the target choices made by women suicide bombers. These researchers underscore the operational advantage that enables women bombers to bypass security measures—they are less likely to raise suspicion and be detected, allowing them to cause more considerable damage to security targets (e.g., Davis 2013; Jacques and Taylor 2008). Their contributions to the area of women suicide terrorism are in line with LaFree and Dugan’s (2009, 417) recommendation that “research would benefit from a more complete understanding of the operational aspects of terrorist behavior.”

However, Galehan’s (2023) histogram reveals a contrasting pattern, showing that women bombers affiliated with Boko Haram tend to target civilians rather than security personnel. Markovic’s (2019, 296) frequency table even demonstrates that women outnumbered men in striking civilian targets. In addition, drawing on a single-country case, Bhatti et al. (2011) and Hsiao-Rei Hicks et al. (2011) provide further evidence of women aiming at civilian populations from their studies on suicide terrorist attacks in Pakistan and suicide bombing casualties in Iraq, respectively.

In recent years, there have been empirical studies focusing on the reasons why women get involved in suicide terrorism. Weeraratne (2022), for example, finds that in Nigeria, female suicide bombers are more likely to target non-religious civilians rather than security targets. Weeraratne’s statistical models consider different types of targets as independent variables. Thomas (2021) demonstrates that female suicide bombers tend to attack and kill civilians more often than other targets. Thomas uses dummy target variables as control variables, not outcome variables. Alakoc (2020)

also reveals that female suicide terrorists are more lethal when targeting civilian populations compared to high-profile individuals and military targets. Alakoc's empirical model specifies the target variable as one of the predictors. Additionally, Soules (2022) explains the scarcity of female suicide bombers despite their operational effectiveness while controlling for the proportion of suicide attacks hitting soft targets such as civilians and political entities. Belgioioso and Eggert (2021) work on civil wars indicates that rebel groups with a higher percentage of female fighters carry out more lethal terrorist operations targeting civilians (see also Fullmer, Mizrahi, and Tomsich 2019; Jahanbani and Willis 2019; Warner and Matfess 2017).

These studies explore the link between suicide attackers' gender and target selection. Yet, they do not explain the reasons behind women's suicide attacks on civilian populations, and they do not provide empirical evidence based on an accurate statistical model that specifies gender as the main predictor and the type of target as the dependent variable.

In this study, I present the strategic and tactical rationale underpinning women suicide terrorism. I argue that terrorist organizations⁴ have a long-term strategy to target civilians and have used women as a tactical tool to cause high death tolls among civilians. To test this argument empirically, I build a model that directly links gender (as a predictor) to different types of targets (as an outcome variable). This approach differs from previous studies. While the literature asserts that women are more likely to attack civilians than soldiers, its statistical model is not capable of determining whether gender directly influences the choice of suicide targets because they include both gender and target variables as causes. I address this issue in my model and examine whether gender significantly impacts the selection of specific target types.

After analyzing 6,127 suicide bombings from 1980 to 2016, my research shows that women are more likely than men to target civilians rather than security objectives. This suggests that many terrorist organizations are using women to carry out attacks on civilian populations, indicating a shift in the traditional roles of women from caregivers to perpetrators of violence.⁵

The Strategic and Tactical Logic of Women Suicide Terrorism

I commence this section with the definitions of key terms, followed by the historical evolution of women suicide terrorism.⁶

In general terms, strategy refers to an overall campaign plan involving complex operational patterns, activities, and decision-making processes that govern tactical execution. Tactics, on the other hand, specifically delineate how organizations plan to achieve a strategic goal. When militant

organizations start a terrorist campaign, they develop a strategy (an overall campaign goal) and tactics (specific and concrete means to achieve that campaign goal). My argument posits that over the past few decades, terrorist organizations such as the LTTE, Taliban, Boko Haram, Hamas, al-Fatah, and al Qaeda have adopted civilian killings as their strategy and employed women as a tactical tool to maximize civilian casualties. Terrorist organizations' primary tactic involves using women to attack civilian populations more often than security targets, though they occasionally dispatch women to strike security targets.

Unlike my argument, several studies maintain that suicide terrorism is intended to bleach security targets. For example, researchers such as Berman and Laitin (2008), Bloom (2005), Crenshaw (2007), Hemmingby (2017), and Piazza (2020) suggest that suicide attacks are effective against well-defended security targets because they are designed to cause severe damage without concern for the attacker's survival (for a dissenting perspective, see Acosta 2016). These studies assert that terrorist organizations are more likely to carry out suicide attacks against heavily fortified military targets, while they tend to use conventional terrorist tactics against civilian targets. Examples of such attacks include Hezbollah's bombing of the U.S. embassy in April 1983 and the assault on United States Marine and French barracks in October 1983 (Lewis 2012). However, it is essential to note that these studies do not consider the gender of the attacker, and they do not address the increased involvement of women in deadly suicide missions targeting civilian populations. By focusing on gender, I explain why women bombers are deployed to strike civilians more frequently than security targets.

Although women have engaged in suicide terrorism since 1985 (e.g., as in the Syrian Social Nationalist Party, the Liberation Tigers of Tamil Eelam, and the Kurdistan Workers' Party), their participation became more active and conspicuous alongside the rise of sectarian violence, largely between Sunnis and Shia. The Sunni versus Shia conflict has provided a springboard for women's involvement in attacking civilian populations. It was initially al-Qaeda in Iraq (AQI)—an Iraqi Sunni Islamic Jihadist organization affiliated with al-Qaeda that later transformed into ISIS—that began inflaming sectarian violence against Shia civilians (Rahimi 2009). In its pursuit to expand its power and influence, AQI adopted a new organizational strategy that focuses on destroying Iraqi Shia civilians instead of security targets, inculcating the idea of civilian killings among members, supporters, and sympathizers (Horowitz 2010).

Abu Musab Zarqawi, the architect of suicide attacks directed at Shia, once declared, "days go by, and events follow one after the other. The battles are many, and the names used are varied. But the goal is one: a Crusader-Rafidite [a derogatory term for Shia] war against the Sunnis"

(Middle East Media Research Institute 2005). Zarqawi's rationale was that if Sunnis could kill enough Shia civilians, the latter would be provoked, resulting in a deeper sectarian conflict. Zarqawi intended to radicalize inattentive Sunnis who would presumably consider their lives in imminent danger stemming from Shia retaliation, even though some are apolitical. AQI expected that more Iraqi Sunnis would join its organization when brutal sectarian violence arose.⁷ This strategy—terrorism with the goal of awakening the masses—shares similarities with historical instances, such as Russian Marxists in the 1900s and leftist groups in Europe and South America in the 1970s, labeling it the “action-reaction cycle.” In their cases, the goal was to provoke a violent response from the incumbent government, revealing its true nature to a quiescent population, whereas AQI purported to incite (civilian) Shia to such acts (Shapiro 2013, 85–89).

When AQI set civilian killings as a campaign strategy, it purported to exaggerate an existential threat to the Sunni community, thereby heightening sectarian violence. Although initially not embraced by other terrorist organizations, the AQI strategy gained momentum among numerous Sunni terrorist organizations and then spread to other organizations that jumped on the bandwagon. Collectively, AQI and other organizations have propagandized a strategy of civilian killings to justify attacking innocent civilians as counterattacks to the ascending power of the Shia. For example, various Sunni terrorist organizations in Lebanon (1975–1990), Iraq (2003–present), Pakistan (2003–present), Nigeria (2009–present), and Syria (2011–present) have launched suicide attacks against Shia populations to incite sectarian violence (Choi and Acosta 2020). Many Sunni militant organizations have recognized these gruesome attacks against civilian targets as a viable option on the pretext of establishing a Sunni version of the ummah in the twenty-first century.⁸ Secular terrorist organizations such as the LTTE also followed suit in targeting more civilians.

When terrorist organizations began targeting civilians, they encountered the challenge of developing specific and effective tactics to achieve this objective. They aimed to inflict mass casualties among civilians. They realized that while men could attack both security and civilian targets, the number of male attackers was limited compared to the vast number of potential targets. Particularly during times of reduced male attacker availability and increased external pressure, these organizations were forced to turn to women as a desperate measure (Nacos 2005; Ness 2005; see also Thomas and Bond 2015; Wood 2019). Consequently, civilian targets were considered less valuable and desirable for men than security targets since they were easier to access and strike. As a result, terrorist organizations began employing male bombers for more complex security targets and female bombers for more easily accessible civilian targets (Gilli and Gilli 2014; Thayer and Hudson 2010). This shortage of male attackers led

to a shift in tactics and influenced the organizations' decisions regarding which targets to pursue.

To achieve their tactical goal of maximizing civilian fatalities, terrorist organizations have increasingly deployed women, as they generally have easier access to targets than men (Berko 2012). This trend is particularly evident in groups with higher numbers of female combatants, who are more likely to target civilians (Harrell 2023).⁹ Bloom (2014) notes a significant shift in the tactics of terrorist organizations: "as terrorist organizations have transitioned from attacking military targets (hard targets) to civilian targets (soft targets), women have proven to be especially useful." This observation underscores how these organizations have turned to using women as a tactical tool to further their goals of civilian killings. This represents a notable change in the dynamics of terrorism tactics among terrorist organizations.

It is essential to understand why targeting security objectives is not advantageous for women. Women bombers typically stand out when attempting to access security targets such as military installations and convoys, as these places are often seen as unsuitable for women due to stereotypical perceptions. For example, women trying to enter a military base are more likely to be stopped and searched because they stand out there than women approaching a market or a restaurant. Similarly, women's attempts to approach security targets traveling through public spaces often raise more suspicion than men who can more convincingly pass as security personnel (Bloom 2014; Davis 2013; Galehan 2023; Jacques and Taylor 2008; Markovic 2019). Meanwhile, male bombers have an advantage in infiltrating these security targets by blending in more effectively, often by disguising themselves as security personnel.¹⁰ By recognizing these differences, I gain insight into why female bombers tend to avoid security targets and instead focus on civilian populations.

As potential targets add more security features in response to the growing threat of terrorist attacks, women prefer to attack civilian targets that have minimal to no security protection. O'Rourke's (2009, 717) analysis alludes to the new pattern of target changes: "if states targeted by female suicide bombers revise their [defensive strategies] to undermine women's advantages in achieving surprise and concealing explosives, I can expect the use of [female suicide terrorists against well-defended, security targets] to decline."¹¹ Bloom (2014) also bolsters this point by indicating an increasing trend in women bombers targeting civilians: "there were changes in targeting. Women are ideal operatives when attacking [civilian], soft targets and blending in with civilians." Similarly, Parashar (2011, 308) maintains that "most targets in Jammu and Kashmir are army camps, police stations, and paramilitary forces. Potential suicide bombers—especially female bombers in civilian clothes—would have difficulty accessing these targets." This

new target trend explains why there have, since 9/11, been more women engaging in suicide attacks against civilians over security targets globally. From a women bomber's perspective, undertaking a deadly bombing mission against civilian targets makes more sense owing to their camouflage advantage—the decreased visibility or attention drawn by a potential woman attacker—than against security targets that are much more difficult to access and destroy.

As women approach civilian targets in public areas—such as malls and funerals, they are less likely to draw the attention of security apparatuses than men. The typical profile of a suicide bomber, often conjured as “a young man, sweating profusely, looking around furtively, carrying a rucksack or wearing bulky clothing” (Edwardes 2005), does not align with women. With this in mind, security personnel are less likely to profile and stop women suicide bombers for body searches in civilian-populated areas. Previous studies indeed point out that security apparatuses often fail to conduct random security checks on women whom they perceive to be nonviolent and, thus, not potential suicide bombers. This lack of suspicion and presumable innocence both provide female bombers with quick and easy access to targets (Sjoberg and Gentry 2007). Accordingly, this presumption leads terrorist organizations to recruit, train, and arm women as ideal combatants for suicide missions after having adopted civilian killings as their strategy and decided to employ women as a tactical tool to maximize civilian casualties.

Traditional attire in regions frequently targeted by suicide bombings, such as the Middle East and South Asia, provides an operational advantage for female bombers, as it allows females to easily conceal suicide belts—explosive devices designed to be strapped to the body—under clothing (Bloom 2011). Faking pregnancy is another useful tool for women who wear suicide belts beneath loose clothing to avoid suspicion and random searches (Bloom 2005, 2014; Cunningham 2007). For example, on March 29, 2002, Ayat al-Akhras, who was then pregnant, detonated her explosives and killed two people at the Kiryat HaYovel supermarket in Jerusalem, Israel (Bennet, 2002b; Israel Ministry of Foreign Affairs 2003). Further leveraging this operational advantage, women holding infants in their arms or walking around with small children appear less suspicious to soldiers and police, reducing the probability of security checks. This operational advantage has even encouraged male bombers to disguise themselves as women to get closer to targets or evade searches by the security apparatus. For example, in July 2017, a suicide attack made by a man dressed in an all-covering robe ended up killing thirteen civilians and one police officer at a refugee camp (Berko 2012; Griffin 2017; O'Rourke 2009). In recent years, younger female attackers have been recruited. Young girls are rarely considered capable of presenting a security threat; therefore, they are prone

to elude terrorist profiling and random security checks. It is difficult for security personnel to imagine a little schoolgirl blowing herself up while pretending to buy noodles from a stall in a popular mall (Warner and Matfess 2017). All these cases illustrate how social biases and the failure of counterterrorism forces to suspect women enable terrorist organizations to utilize female bombers who have easier access to targets and cause large death tolls (Cunningham 2007; Davis 2013; Galehan 2023; O'Rourke 2009; Speckhard 2008).

The distinct advantage of female suicide bombers lies in their ability to blend in more effectively and pass through security with less scrutiny compared to their male counterparts. This often results in a surprising and devastating impact on their attacks (Davis 2013; Jacques and Taylor 2008; O'Rourke 2009; Sjoberg and Gentry 2007). In 2017, Boko Haram used women as suicide bombers more than any other terrorist organization. The interception rate for female bombers was lower than that for men, allowing for easier access to their intended targets (Global Extremism Monitor 2018; Pearson 2018; Warner and Matfess 2017). However, on some occasions, security services may be on high alert for women, making it harder for them to reach their targets. After Imarat Kavkaz employed female attackers intending to expel the Russian presence from the North Caucasus, the Russian Federal Security Service started conducting security checks on women.¹² However, stopping every female suspect for security has remained challenging in open civilian spaces like malls and parks as they do not stand out there.¹³ In cultures where it is deemed socially unacceptable for men to conduct body searches of women, security challenges are likely to arise. Paradoxically, some women use beauty to avoid suspicion or detection (Bloom 2011). For instance, Leila Khaled, a member of the Popular Front for the Liberation of Palestine, shared her experience with a male security guard: "I had a pistol in my belt, a grenade in my pocket, and TNT in my bag. I was a woman dressed fashionably. I opened my bag for security, but the man just saw my make-up and waved me through" (Kendall 2015). This news story illustrates that even when women are subjected to extra security scrutiny, they are still better positioned to bypass security checkpoints than men.

It is worth noting that when terrorist organizations decide to carry out a suicide attack, they often involve women in the decision-making processes. These organizations recognize that women may prefer attacking civilians instead of security targets due to easier access, and they are willing to risk their lives in the process. This observation aligns with Wood's (2019) point that "even though the leadership retains ultimate discretion over women's roles, they sometimes succeed in advancing their collective interests within the organization" (see also Thomas and Bond 2015, regarding the organization's demand for women). By listening to

women, terrorist organizations can better motivate them and enhance their commitment to the suicide mission. By giving women an opportunity to voice their preferred target, terrorist organizations can reassure them that they are not suicide robots but God's instruments—he seeks to use them for his purpose to bring glory to his name, and they are also their invaluable tactical assets to fulfill organizational goals if they are not religious (e.g., female Tamil Tigers).

The following study illustrates that targeting civilians is not a unilateral decision of terrorist organizations but the two-way communication between organizations and individual bombers:

Although the cell members were aware of the importance of achieving high media impact for their actions, when it came to choosing targets, operational viability took precedence over the potential symbolism of an attack. For instance, when Antonio sounded his colleagues out as to whether they would be willing to behead an “infidel” who was already under surveillance, they expressed an interest in knowing the identity of the target. Asked if the person was a banker or a company boss, Antonio ruled out such a possibility given that “those people are Swiss bankers and Jews, and have bodyguards and all that, so it is not easy to kill them” (Torres-Soriano 2021, 1369).¹⁴

The testimonial of a LTTE woman is also telling:

Female (and male) terrorists and guerrillas often show rationality in their decision to take up arms. As one LTTE woman stated, “I’ve made my choice” (Stack-O’Connor 2007, 58–59).

It is not uncommon for women terrorists to have target preferences. Because women are familiar with civilian sectors, they can approach civilian populations without being detected and caught. Their confidence in easier access to civilian populations is likely to create the amount of damage that is set by their organization’s strategic goal of civilian killings. Numerous cases show that women bombers expressed their preferences for targeting civilians over security targets. For example, Berko (2012, 74), who analyzes the target choices of Palestinian terrorists, states: “many male terrorists I interviewed claimed they didn’t want to kill Israeli civilians. Most of the women, on the other hand, told me enthusiastically that they preferred killing civilians” (see also Bloom 2014; Stack-O’Connor 2007). Several studies offer additional evidence for the bombing choices of women who are more likely to go after civilian targets than men (e.g., Bhatti et al. 2011; Hsiao-Rei Hicks et al. 2011). An illustrative example is the case of Wafa Idris, a 28-year-old woman who detonated a 22-pound bomb in the center of Jerusalem outside a shoe store on Jaffa Road in 2002. Her suicide bombing was seen as “a sign of the growing desperation among Palestinians, and the growing willingness of Palestinian militants to deploy women who can slip past Israeli security” (Bennet 2002a). The Aqsa Martyrs’ Brigades,

which was responsible for the bombing, included Wafa Idris in the target selection, as she had an independent mind and made her own decision. She became the first female suicide bomber in the Israeli-Palestinian conflict, and since then, other Palestinian women have followed her lead (Speckhard 2008). Over the past few decades, many women around the world have increasingly engaged in suicide terrorism, targeting civilian populations instead of security targets.

Overall, the current literature neglects to address the critical role of women in suicide terrorism adequately, failing to consider gender differences when examining the relationship between suicide bombers and their choice of targets. After conceptualizing that women are used as a tactical tool in suicide attacks to achieve the strategic goals of the terrorist organization, I argue that women suicide terrorists are more commonly used to target civilians rather than security targets because they can more easily blend in or disguise themselves.

H₁: Women are less likely to carry out suicide bombings against security targets than men.

Five Alternative Explanations That May Affect Women Suicide Bombers and Security Target

To my knowledge, no studies currently examine the likelihood of female suicide bombers attacking civilians over security targets. Therefore, it is challenging to find other factors that may affect both explanatory and outcome variables.¹⁵

Based on my literature review, I have selected five variables that may influence the choice of targets in suicide terrorism. These variables are religious conflict, the involvement of Islamist terrorist organizations, transnational terrorist organizations, foreign occupation, and unique political and security conditions in Iraq and Syria. The religious conflict variable assesses the extent to which religion influences the selection of targets for suicide attacks. The variables related to Islamist and transnational terrorist organizations evaluate whether the organizational characteristics of these groups affect their choice of targeting security forces or civilian entities. Lastly, the controls of foreign occupation and the Iraq and Syria variable help assess the impact of the host country's environment on the outcome.

Religious Conflict

Numerous studies suggest that both civil and international wars lead countries to strengthen the protection of potential targets, especially key security interests such as military personnel, equipment, important political

figures, and government buildings (Carter 2012; Piazza 2020; Thomas 2021). During religious conflicts, countries are likely to further bolster protection around these security targets due to their severity and brutality (Berko 2012).

H₂: Religious conflicts are more likely to engender suicide bombings against security targets than non-religious conflicts.

Islamist Terrorist Organization

Islamist terrorist organizations aim to establish a global caliphate or to demonstrate organizational loyalty to this cause (Asal et al. 2009; Berman and Laitin 2008). These organizations frequently carry out suicide bombings targeting security objectives. My data shows that Islamist terrorist organizations account for 90 percent of these attacks. This finding aligns with Choi and Brown (2022) research, which suggests a positive association between Islam and the occurrence of suicide attacks.

H₃: Islamist terrorist organizations are more likely to carry out suicide bombings against security targets than non-Islamist terrorist organizations.

Transnational Terrorist Organization

Another potential factor to consider is the size of terrorist organizations. Specifically, it is important to determine whether their activities are confined within the borders of a single state (Choi 2024; Choi and Acosta 2020). Transnational terrorist organizations that operate across borders tend to have access to more resources, both material and financial, compared to domestic organizations. This access enables them to more effectively target security interests, enhancing their perceived strength and reputation.

H₄: Transnational terrorist organizations are more likely to carry out suicide bombings against security targets than non-transnational terrorist organizations.

Foreign Occupation

Foreign military forces occupying a country can foster resentment among the local population, potentially leading to increased attacks on military personnel and equipment, particularly those belonging to foreign powers (Choi and James 2016; Choi and Piazza 2017; Danzell, Kisangani, and Pickering 2019; Pape 2005).

H₅: Countries under foreign occupation are more likely to provoke suicide bombings against security targets than those not under foreign occupation.

Iraq and Syria

Another factor is the unique political and security environment of Iraq and Syria, two countries most prone to terrorism. This factor may significantly influence the decisions made by terrorist organizations and their operatives regarding the selection of security versus civilian targets. It should be noted that there was a surge in Sunni suicide attacks around 2006–2007, coinciding with a peak in sectarian violence in Iraq. A second spike occurred during 2014–2015, which coincided with a rise in attacks by the Islamic State in both Iraq and Syria (Choi and Acosta 2020).

H₆: Iraq and Syria are more likely to experience suicide bombings against security targets than other countries.

Research Design

To empirically test the above hypotheses, I gather available information on suicide bombing incidents, including the gender of the terrorist. My data is sourced from the Suicide-Attack Network Database (SAND), accessible at <https://www.revolutionarymilitant.org/sand>,¹⁶ and other credible online sources. The SAND includes 6,224 suicide bombing incidents recorded from 1980 to 2016, excluding failed bombings with no explosion. I compile a sample of 59 countries in which suicide incidents occurred over the past four decades. This sample of countries ensures that my estimates are not biased by including nations with no history of suicide terrorism. Since my theoretical discussion focuses on how gender affects the type of target, I organize the dataset at an individual target level (i.e., security versus civilian targets) and subsequently construct a statistical model as follows:

$$\begin{aligned} \text{Security Target} = & \alpha_0 + \beta_1 * \text{Female Suicide Bomber} + \beta_2 * \text{Religious Conflict} \\ & + \beta_3 * \text{Islamist Terrorist Organization} \\ & + \beta_4 * \text{Transnational Terrorist Organization} \\ & + \beta_5 * \text{Foreign Occupation} \\ & + \beta_6 * \text{Dummy for Iraq and Syria} + \varepsilon \end{aligned}$$

The dependent variable, Security Target, is dichotomous, coded as “1” when the terrorist target is security-related (e.g., military soldiers or police station) and “0” for civilian targets (e.g., unarmed individuals or schools). Unlike security targets, civilian targets typically lack easily recognizable security features such as gates, guns, and guards. I operationalize the Security Target variable based on the description of the type of target (i.e., the TARGET variable) in the SAND and on reporting in various media outlets. Due to a lack of clear or authoritative definitions of security

versus civilian targets in the literature, I follow Piazza's (2020) coding scheme, classifying terrorist targets into six types: military, police, government, rebel, civilian, and unknown. The military target category includes military soldiers, equipment, facilities, and checkpoints; the police target category contains police forces, police stations, and security points; the government target category covers prominent political figures such as presidents, governors, and securitized government facilities; the rebel target category consists of non-state actors such as militant rebels and terrorist organizations; the civilian category encompasses unarmed people, schools,¹⁷ markets, and religious centers; and the unknown target category denotes suicide bombings that were prematurely detonated before arriving at their unknown destinations.

Out of the 6,224 suicide bombing incidents, I categorize 2,035 (33%) as military targets; 1,257 (20%) as police targets; 657 (11%) as government targets; 272 (4%) as rebel targets; 1,906 (31%) as civilian targets; and 97 (2%) as unknown. Since unknown observations are not useful for my classification of security versus civilian targets, I exclude them, leading to a sample composed of 6,127 suicide bombing incidents.

Based on these five categories, I create my dependent variable of Security Target. I dichotomize the variable, denoting "1" for any suicide bombing targeting the military, police, or government and "0" for rebels and civilians.¹⁸ The rationale behind categorizing rebel groups as civilian targets is that they often engage in asymmetric warfare, where their armed strength is weaker than that of the government military forces (Thornton 2007), aligning them more with the characteristics of civilian targets. To ensure the validity of categorizing rebel groups, I conduct two checks. First, I remove rebels from the Civilian Target category and classify them as part of the Security Target. As shown in [Supplementary Appendix 2](#), this modified categorization does not meaningfully change the main finding of this study: female bombers carry out suicide bombings against security and rebel targets less frequently than their male counterparts. Second, following Polo's (2020) approach, I exclude rebels from my sample data. The results, reported in [Supplementary Appendix 3](#), confirm that female bombers are less likely to go after security targets. As the rebel category constitutes only about 4 percent of the sample observations, its effect is negligible in the data analysis, regardless of the operationalization. These checks reinforce the robustness of my analysis and indicate that irrespective of how rebel groups are categorized, the main conclusion about the lower likelihood of female bombers targeting security holds true.

Among 6,127 suicide bombings, security targets comprise 3,949 (64%), while civilian targets comprise 2,178 (36%). This percentage difference is consistent with the observations made in existing studies. For instance, Berman and Laitin (2008) note that suicide terrorism primarily aims to

bleach security targets. Figure 1 displays the graphic pattern of security versus civilian targets over the last four decades. Despite suicide bombings typically targeting security-related entities, the increasing number of incidents involving civilian targets over time is intriguing. The figure provides clear evidence of changes in the strategic and tactical thinking of terrorist organizations and their bombers. Ultimately, this upward trend of civilian targeting poses a new puzzle for scholars studying the behavior of individual suicide bombers in target selection.

As the dependent variable, Security Target, is dichotomous, I employ standard logistic regression for estimation. It is possible that women bombers initially targeted security locations, but later switched their focus to civilians as security targets became better protected. In other words, as the global landscape of women suicide bombers has changed over the years, the effect of time should be considered in statistical modeling. To account for this, I incorporate a temporal element into the model specification by including year fixed-effects.¹⁹ To address concerns about potential model misspecifications, my models report coefficients with robust standard errors. For added reliability, I also consider the clustering of sample data—security targets are clustered in certain countries with more resources to bolster their security environment than others.²⁰ Such clustered data may require a conservative estimation approach, reporting more nuanced standard errors that address intra-country correlation. Additionally, gender and target choices may be endogenous: while gender determines the type of target, a certain target might also affect the choice of women suicide bombers. As indicated by Soules (2022), who studies the determinants of gender, I use an instrumental

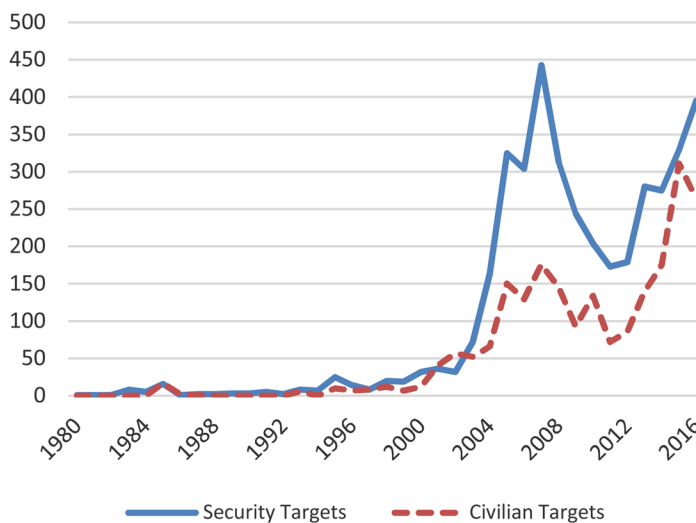


Figure 1. Security versus civilian targets.

variables probit model to mitigate potential endogeneity bias, as described in the next section.

The main independent variable, Female Suicide Bomber, captures whether the attack includes a terrorist who is believed to be female. I code it as “1” when the perpetrator was a female in the bombing incident and “0” for a male.²¹ To determine the gender of a suicide bomber, I refer to various online media outlets in addition to SAND data. Given that female suicide bombers are likely to receive more media attention than their male counterparts (Avraham 2016), I find more gender-related information through online media. When I encounter instances of a man disguising himself as female-presenting, I place him in the female category. This decision is based on his utilization of the tactical advantages of being female-presenting, aligning with the focus of my study on the role of gender in carrying out an attack rather than biological sex. It is important to note that even after re-coding the cases of men disguised as women to men, the main results remain the same, mainly due to the rare instances where men used such tactics, leading to a negligible impact. When an attacker’s gender is unknown, I code it as male since male bombers often receive less media coverage,²² following the common practice in existing studies, such as in Alakoc (2020). It is worth noting that women suicide bombers make up a small portion—302 (5%)—of the total 6,127 suicide bombing incidents,²³ with the majority involving male suicide bombers, totaling 5,825 (95%).

Religious Conflict: The variable is dichotomous, coded as “1” when suicide attacks occurred in the context of religious conflict and “0” when there was no religious conflict. The data for this variable is derived from the SAND database.

Islamist Terrorist Organization: I code this variable as “1” if Islamist organizations carried out a suicide attack and “0” otherwise. The data for this variable is collected from the SAND.

Transnational Terrorist Organization: This variable is also a dummy variable, coded as “1” if transnational terrorist organizations executed a suicide attack and “0” if they did not. This data is gathered from the SAND.

Foreign Occupation: The foreign military occupation variable is dichotomous, coded as “1” for observations where the country was under foreign military control and “0” for non-events. This variable is based on the category of –66 in the composite democracy score of Polity, indicating cases of foreign “interruption” (Marshall, Gurr, and Jagers 2017).

Dummy variable for Iraq and Syria: I create a dummy variable for Iraq and Syria, which are two highly contentious countries where suicide terrorism has occurred most frequently.

Table 1 displays the descriptive statistics for all the variables discussed above.

Table 1. Descriptive statistics.

Variable	Obs	Mean	Std. Dev.	Min	Max
Security target	6,127	0.645	0.479	0	1
Female suicide bomber	6,127	0.049	0.216	0	1
Religious conflict	6,127	0.929	0.257	0	1
Islamist Terrorist Organization	6,127	0.903	0.296	0	1
Transnational Terrorist Organization	6,127	0.447	0.497	0	1
Foreign occupation	6,127	0.403	0.490	0	1
Dummy for Iraq and Syria	6,127	0.434	0.496	0	1

Empirical Results

Table 2 presents the estimated logistic regression results, both with and without clustering.²⁴ The overall findings support my main hypothesis about gender differences. Whether clustering is implemented or not, female suicide bombers breach security targets less frequently than their male counterparts.

To ensure the validity of my findings, I employ four different model specifications: Models 1 through 4 take a step-by-step modeling approach from bivariate to multivariate regression. Model 1 tests the relationship between gender and target in a parsimonious way; Model 2 adds the religious factor; Model 3 includes two additional variables related to organizational characteristics; and Model 4 presents a full model specification with two more variables that control for country characteristics. The coefficient on the Female Suicide Bomber variable across the four models is significant at the 0.001 level and in the expected direction. This consistent result highlights the crucial role of gender in suicide terrorism, indicating that women are less likely to attack security targets than men. This finding further suggests that women's suicide attacks on civilian targets increasingly replace conventional terrorist attacks previously primarily carried out by male perpetrators. Accordingly, my empirical analysis offers a critical departure from previous studies of suicide terrorism and points toward a new direction of inquiry.

The sample data are organized at more than one level, with security targets clustered in countries. This may call for specifying another type of standard error to address the following data structure: the choice of a security over a civilian target varies depending on the country of residence and the gender difference. To account for clustering, I incorporate intra-group correlation into the estimation. Consistent with the previous four models, the re-estimated results in Models 5 to 8 in **Table 2** show the Female Suicide Bomber variable as a significant and negative predictor. This cluster modeling reaffirms the main hypothesis regarding the different roles of female versus male bombers.

Table 2. Do female suicide bombers go after security targets?

Variable	Logit: Not clustered				Logit: Clustered			
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Gender difference								
Female suicide bomber	−0.884*** (0.123)	−0.880*** (0.123)	−0.904*** (0.125)	−0.879*** (0.127)	−0.884** (0.274)	−0.880** (0.279)	−0.904** (0.297)	−0.879** (0.303)
Religious conflict								
Religious conflict		0.190 (0.111)	0.633*** (0.151)	0.615*** (0.150)		0.190 (0.455)	0.633 (0.407)	0.615 (0.390)
Organizational characteristics								
Islamist			−0.485*** (0.143)	−0.473** (0.146)			−0.485 (0.333)	−0.473 (0.335)
Terrorist Organization								
Transnational Terrorist Organization			−0.271*** (0.060)	−0.321*** (0.091)			−0.271 (0.311)	−0.321 (0.276)
Country characteristics								
Foreign occupation				0.674*** (0.085)				0.674** (0.222)
Dummy for Iraq and Syria				0.035 (0.093)				0.035 (0.183)
Constant	0.528*** (0.079)	0.346** (0.133)	0.583*** (0.142)	0.602*** (0.150)	0.528*** (0.134)	0.346 (0.498)	0.583 (0.461)	0.602 (0.403)
Number of countries	na	na	na	na	59	59	59	59
Pseudo R^2	0.03	0.03	0.03	0.04	0.03	0.03	0.03	0.04
Observations	6,127	6,127	6,127	6,127	6,127	6,127	6,127	6,127

Note: Robust standard errors in parentheses.

* $p < .05$; ** $p < .01$; *** $p < .001$, two-tailed tests.

Among the seven control variables in Models 1 through 8, Foreign Occupation is the only variable that consistently achieves significance. When foreign powers invade and occupy a country, there appears to be a higher incidence of suicide bombings directed toward security targets, mainly pertaining to foreign militaries.

To visualize the estimated coefficients, I plot the average marginal effects of all predictors in Figure 2 based on the log odds of Model 8. These average marginal effects provide insights into the average probability change as each of the six predictors increases by one unit (e.g., when the suicide bomber changes from male to female bomber). The x-axis indicates the average marginal effects: the increased or decreased magnitudes evaluated by the position and length of the bar. This visual analysis confirms that female suicide bombers are less likely to strike security targets.²⁵ In short, the analysis of substantive effects aligns with the results of the conventional significance-level tests.

Table 2 benefits from parsimonious models that aim to explain data using a minimal number of predictor variables. This approach helps me avoid including too many controls, in line with Achen's (2005, 327) caution that "dumping long lists of explanatory variables into linear regression... [can lead to biased estimates]." In his earlier work, Achen (2002, 446) even suggests a rule of three: "a statistical specification with more than three explanatory variables is meaningless." However, it is still valuable to

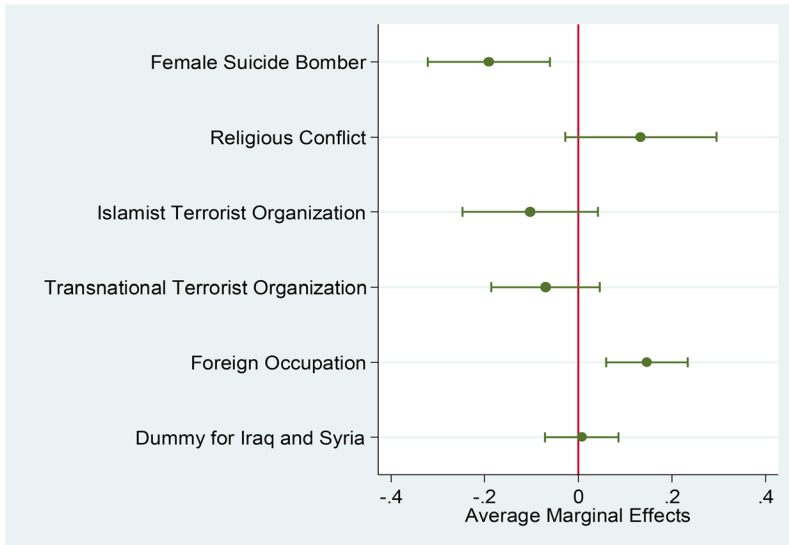


Figure 2. Average marginal effects.

examine whether a third factor influences the relationship between women suicide bombers and security targets. To be thorough, I have added five variables: forced recruitment, leftist ideology, group size, conflict duration, and the target state's level of democracy.

Some terrorist organizations resort to abduction, threats, or the use of force to conscript recruits within their constituencies (Berman 2004; Eck 2014; Tokdemir and Akcinaroglu 2016). For instance, Boko Haram kidnapped schoolgirls and individuals with mental disabilities. It then forced them to wear explosive belts and position them near potential targets. However, many of these girls often abandoned their tasks due to fear, resulting in little to no damage to the intended targets (Pearson 2018; Searcey 2017; Warner and Matfess 2017).

When terrorist organizations forcibly recruit their members, they are less likely to grant autonomy to women. Consequently, they are more prone to committing violence against civilians rather than security targets, as civilians are more manageable to infiltrate. I create a forced recruitment variable by consulting Tokdemir and Akcinaroglu's (2016) Reputation of Terror Groups (RTG) Dataset,²⁶ the Global Terrorism Database (GTD),²⁷ Acosta's (2019) Revolutionary and Militant Organizations dataset (REVMOD),²⁸ and various open media sources. I record this variable as "1" if a terrorist organization is known for forced recruitment and "0" otherwise.

Leftist ideologies aim to promote greater equality and to reduce or eliminate privilege, hierarchy, and gender disparities (Lutz and Lutz 2008). Unsurprisingly, the New Left in the U.S. played a significant role in inspiring women's liberation during the civil rights movement (Evans 1979). I

conceptualize that terrorist organizations that advocate leftist ideologies are more likely to recruit women, as they view women as equally capable, both intellectually and physically. For instance, the Revolutionary Armed Forces of Colombia (FARC) have included women in various roles, including combat positions, which provided them with a sense of purpose, fulfillment, and equality (Leech 2011). I expect that left-wing terrorist organizations tend to commit lower levels of violence against civilians, as their primary targets are the states they oppose. I categorize left-wing terrorist organizations as “1” and all others as “0.” I collect data from sources such as Acosta’s (2019) REVMOD, the GTD, Harrell (2023), Tokdemir and Akcinaroglu’s (2016) RTG Dataset, and various open mass media outlets.

When testing the effect of political command²⁹ on organizational success, Acosta (2019) specifies group size, conflict duration, and the target state’s democracy level as control variables. Acosta reasons that organizational size boosts prospects for success. I theorize that as terrorist organizations grow bigger, they become more capable of attacking security targets since they will have more material resources and manpower. Acosta conceptualizes that conflict duration has an inverse relationship with success. I argue that as terrorist organizations have a long-lasting conflict with the government, they are more likely to encounter soldiers and police officers who more vigorously go after terrorists in the hope of ending the continued conflict. Acosta finds no significant relationship between political command and the target state’s democracy level. I expect that as political leadership becomes better organized and united for all operations, it is more likely to plan and execute attacks on security targets since it will gain more organizational strength (see Choi 2010, 2016). I garner data on group size, conflict duration, and the target state’s democracy level from Acosta’s (2019) REVMOD and other public sources such as Polity IV.

In [Supplementary Appendix 6](#), Models 1 and 2 present the estimated coefficients and standard errors obtained from replicating the last model in [Table 2](#) after adding forced recruitment, leftist ideologies, group size, conflict duration, and the target state’s democracy level. The replicated results are consistent with the main findings of this study. Model 1 shows a significant, negative effect of Female Suicide Bombers, even after controlling for Forced Recruitment. The Leftist variable is excluded from the model due to collinearity. Model 2 turns out that women are still more likely to target civilians while controlling for factors such as group size, conflict duration, and the target state’s level of democracy.

It is intriguing to evaluate whether the main empirical findings hold when using widely used terrorism datasets, such as the Database on Suicide Attacks (DSAT) from the Chicago Project on Security and Threats (CPOST) and the Global Terrorism Database (GTD), instead of the SAND. The

SAND is known for its comprehensive coverage of suicide bombing incidents, including details about the gender of attackers (Acosta 2016). In contrast, the GTD lacks gender-related information, which is essential for understanding the complexities of suicide attacks. The DSAT from CPOST, however, provides information about the perpetrator's gender.

I re-estimate the last four models in Table 2 by swapping the gender-related variables from the SAND dataset with those from the DSAT dataset. To gather gender-related variables, I reference Thomas (2021) study, which conducted a statistical analysis using data on suicide attacks from the CPOST. Her study has already undergone a rigorous peer review process in a premier journal. By utilizing Thomas' replication materials, I aim to minimize any biases that could inadvertently arise from my own data collection from the CPOST's DSAT. I present the replicated results in Models 3 to 6 in Supplementary Appendix 6. Overall, the findings across these models again confirm that female suicide bombers are less likely to breach security targets.

One may contemplate that terrorist organizations make a tactical decision to deploy women against civilian targets and men against security targets since the former are better at striking civilian targets due to their innocent looks and blending with non-security people than the latter. At the same time, it is possible that even military installations in highly patriarchal countries have women serving in support roles, or they may be present as wives or prostitutes. These two dynamics might be more pronounced in countries with lower levels of gender equality. In her 2021 research, Thomas develops and tests this idea as her second hypothesis: "suicide attacks committed by female terrorists are more lethal than those committed by male terrorists in societies with more conservative gender norms" (p. 779). Her first hypothesis states that "suicide attacks committed by female terrorists are more lethal than those committed by male terrorists" (p. 777). It is important to note that both hypotheses focus on the impact of gender on the greater lethality of suicide attacks carried out by women.

My study focuses on women who engage in attacks against civilians, specifically examining the role of female suicide bombers without exploring the interaction between women and gender equality. I expect that there may be fewer attacks by women in non-patriarchal societies, which aligns with Huber's (2019) argument that as gender equality improves, the number of attacks on civilians, relative to those on security targets, decreases. Due to space constraints, I will leave a comprehensive theoretical discussion of this interaction effect for future research, but I conduct preliminary statistical testing using a multiplicative regression model.

I introduce a gender equality variable, proxied by women's civil society participation as in Thomas (2021) study,³⁰ and its interaction

term with Female Suicide Bomber to the last models of [Supplementary Appendix 6](#).³¹ [Supplementary Appendix 7](#) presents the estimated results obtained from the multiplicative logit regression model. Interpreting coefficients in [Supplementary Appendix 7](#) is challenging due to the non-linear nature of the interaction terms, so I turn to graphic analysis. I plot the average marginal effects of the interaction variable, which appears in full Model 6 of [Supplementary Appendix 7](#). I draw [Supplementary Appendix Figure 2](#) based on the log odds of the multiplicative logit model. This figure shows evidence that female suicide terrorists are less likely to target security forces compared to their male counterparts in societies with less conservative gender norms.

Distinguishing between security and civilian targets may lack complete clarity, as researchers sometimes hold differing opinions on how to precisely identify and categorize them (Piazza 2020). To address this issue, I provide five different robustness tests in [Table 3](#).³² These robustness tests further bolster my main hypothesis concerning gender differences across Models 1 through 5, even with the instrumental approach. Below, I explain how each robustness test is performed.

Table 3. Do female suicide bombers go after security targets? Robustness tests.

Variable	Logit: clustered				Probit: instrumental variables	
	Military target	Combatant target	Foreign target	Shia civilians	Female bomber (first stage)	Hard target (second stage)
	Model 1	Model 2	Model 3	Model 4	Model 5	
Gender difference						
Female suicide bomber	−0.924*** (0.253)	−0.693** (0.213)	−1.281** (0.454)	1.222*** (0.173)		−1.768 (0.927)
Religious conflict						
Religious conflict	0.578 (0.422)	0.035 (0.273)	−0.105 (0.261)	2.267*** (0.683)	−0.068 (0.045)	0.277 (0.279)
Organizational characteristics						
Islamist Terrorist Organization	−0.453 (0.333)	−0.221 (0.300)	−0.209 (0.409)	1.967 (1.183)	0.049 (0.045)	−0.246 (0.220)
Transnational Terrorist Organization	−0.328 (0.217)	−0.297 (0.220)	0.264 (0.341)	0.333 (0.311)	−0.008 (0.044)	−0.149 (0.150)
Country characteristics						
Foreign occupation	0.349 (0.211)	0.658** (0.201)	1.920*** (0.406)	−0.704** (0.265)	−0.021 (0.033)	0.347* (0.138)
Dummy for Iraq and Syria	0.169 (0.239)	0.398 (0.218)	−1.898*** (0.245)	−0.218 (0.253)	0.002 (0.040)	−0.043 (0.105)
Instrument						
Terrorist competition					−0.008* (0.004)	
Constant	−0.100 (0.650)	0.971* (0.382)	−1.741** (0.590)	−5.520*** (1.151)	0.181*** (0.051)	0.366* (0.185)
Number of Countries	59	59	59	59	59	59
Pseudo R^2	0.03	0.04	0.14	0.09	na	na
Observations	6,127	6,127	6,127	6,127	6,127	6,127

Note: Robust standard errors in parentheses.

* $p < .05$; ** $p < .01$; *** $p < .001$, two-tailed tests.

In Model 1, I narrow the type of security targets to military only, as military soldiers, equipment, and bases should be more secure from suicide attacks than any other potential targets. In Model 2, I select terrorist targets involving combatants only because protecting fighters during a conflict should be a priority. In Model 3, I choose targets located outside of the national territory, or that belong to foreign entities in the venue country on the premise that foreign powers should possess strong military prowess and resources. In Model 4, I examine the effect of Sunni women bombers attacking Shia civilians only. I expect a positive relationship since Sunni terrorist organizations have frequently engaged in sectarian violence against Shia civilians.

Model 5 provides an analysis of reverse causality. My theory predicts one-way causality going from Female Suicide Bomber to Security Target; therefore, the empirical models are designed to test the one-way causal relationship. However, there is a possibility that while the gender difference affects the type of target, a certain target might influence the choice of female suicide bombers, as implied by Soules (2022), whose dependent variable is gender. Though it is beyond the scope of this study, I introduce a simultaneous equations model to account for endogeneity. For Model 5, I return to the original measure of the dependent variable—Security Target—but employ a simultaneous equations estimation strategy.

I run an instrumental variables probit model to address the issue of endogeneity.³³ Identifying an ideal instrumental variable is quite challenging. As no alternatives have been suggested in the literature, I use competition among terrorist organizations as my instrumental variable. This choice broadly aligns with the perspective of Thomas and Bond (2015) who state: “especially as they compete with other organizations for new members, prestige, and public support, [terrorist organizations are willing] to lower the costs of participation for women volunteers...[as] women participants can provide, particularly with respect to productivity and tactical efficiency.” I expect that when terrorist organizations compete for public support, they are less likely to deploy women as bombers (see Choi 2018). This is because competing organizations typically send male bombers to attack well-defended security targets to outdo one another, while they tend to dispatch female bombers to strike civilian targets, where they can blend in more easily (Bloom 2005). Successful attacks on security targets will likely grant the organization more recruits, prestige, and public support.

To ensure that competition among terrorist organizations meets the exclusion restriction for an instrumental variable, I utilized a statistical test developed by D’Haultfœuille, Hoderlein, and Sasaki (2021, 2024), and it passed the test. I measured competition among terrorist organizations by counting the number of rival organizations in each country, based on the SAND dataset.

Model 5 displays the estimated results obtained from the simultaneous equations regression. In the left column under Model 5, coefficients and standard errors produced by the first-stage regression are displayed, while the right column shows those from the second-stage regression. The estimated results indicate the important role of gender with a p -value of 0.057.

Conclusion

Various terrorist organizations, including the Tamil Tigers, Hamas, al-Fatah, Boko Haram, Chechen organizations, and several al-Qaeda affiliates, have extensively used women in their suicide attacks. However, existing studies overlook strategic and tactical explanations for why women target civilian populations. Additionally, while some studies claim to find empirical evidence of women's preference for attacking civilians over soldiers, their statistical models are not suitable for drawing this conclusion as they include both gender and target types as predictors on the right-hand side of the equation. In this study, I conceptualize the reasons behind women's preference for targeting civilian populations and then conduct empirical research by directly examining the impact of women (the main predictor) on a specific target type (the dependent variable). This is my attempt to address Crenshaw's (2007) criticism, where scholars rarely consider variation in the target set (e.g., civilian, security, political, etc.).

I discover strong and consistent evidence indicating that women are increasingly carrying out suicide attacks targeting civilian populations. This shift toward civilian targets remains pronounced even when controlling for potential shifts in target security measures (e.g., foreign occupation), group objectives (e.g., Islamist terrorist organizations), changing resource access, and recruitment capabilities of groups (e.g., transnational terrorist organizations). This emerging trend is noticeable in the context of sectarian conflict. It seems that Sunni terrorist organizations are using women bombers to provoke sectarian violence as part of their efforts to regain their former influence (see Model 4 in Table 3).

These findings offer new insight into the role of women in suicide terrorist attacks. A global trend has emerged where women terrorists are increasingly targeting civilian populations in public spaces, rather than focusing on security targets in restricted areas. Consequently, the counterterrorism community must develop effective strategies to address the growing security threats posed by women suicide bombers.

Notes

1. Many scholars define terrorism as "the threatened or actual use of illegal force, directed against civilian targets, by nonstate actors, to achieve a political goal through

fear, coercion, or intimidation” (LaFree and Ackerman 2009, 348; see also Hoffman 2017). In this study, however, I adopt a broader definition of terrorism, which encompasses political violence against both civilians and military personnel. I use the terms terrorism and terrorist attacks interchangeably.

Suicide terrorism refers to “an act of political violence that begins with the attempted—usually successful—suicide of its perpetrator” (Acosta 2016, 180), intended to harm or kill civilians and military personnel. For the sake of analytical simplicity, I use the terms suicide terrorism, suicide attacks, and suicide bombings interchangeably.

2. In scholarly literature, security and civilian targets are often categorized as hard and soft targets (e.g., Piazza 2020). For a general readership, I use the former terms throughout this study.
3. While not specific to suicide terrorism, Polo (2020) offers a comprehensive explanation of terrorist organizations’ target preferences, examining how a group’s ties to its constituency and specific government repressive strategies can either constrain or incentivize terrorist attacks against soft civilian versus security targets. Marchment and Gill (2020) propose another explanation rooted in spatial decision-making: targets are selected based on being tolerable, relevant, accessible, close, and/or known (see also Asal et al. 2012; Brandsch and Python 2021; Valentino 2014).
4. In this study, I borrow the definition of Hou, Gaibullov, and Sandler’s (2020, 201) terrorist organization—“a subnational collective whose members try to obtain a political goal by intimidating an audience (or constituency) through violence or its threat.” See Phillips (2015) and Young and Shellman (2019) for more nuanced definitions.
5. It is worth noting that relying on Loken and Matfess (2024) Women’s Activities in Armed Rebellion (WAAR) project, Harrell (2023) sheds new light on women’s role as frontline fighters versus non-combat outreach.
6. On the causes and effects of suicide terrorism, see Choi 2015, Choi 2019; Choi and Piazza 2016; Pape 2005.
7. There was disagreement over attacking Shia civilians even within AQI. Ayman al-Zawahiri expressed concern that AQI was alienating Sunnis in Iraq by engaging in civilian killings (Cordesman 2006; Engel 2015).
8. Most sectarian violence occurred between Sunni and Shia Muslims. However, there have also been cases of Shia suicide attacks against Sunni mosques, with each attack typically triggering a response from the other side.
9. However, Mehr’s (2023) case study on the Nepalese Civil War indicates that “all-male rebel units were responsible for 25% more killings of civilians than perfectly balanced ones.”
10. If both men and women disguise themselves as security support staff, such as cleaners and cooks, gender differences should be blurred.
11. Ten years later, Fullmer, Mizrahi, and Tomsich (2019, 277) also observe that “following the increase in the frequency of female attacks in 2008, and the particularly lethal attacks during 2009 and 2010, security forces may have adjusted their tactics, resulting in proactive responses to the threat of female-perpetrated suicide bombings.” A conclusion can be drawn from this observation: the benefits of women suicide bombings has declined over time. Thomas (2021, 793–795) tests confirm the conclusion.
12. See <https://www.globalsecurity.org/military/world/para/ik.htm>.
13. In contrast, men are likely to be searched if they fit the typical profile of a male suicide bomber.

14. One may argue that the statement about women showing “an interest in knowing the identity of the target” does not imply that they refused to attack military targets or explicitly requested not to do so. However, the suicide bomber was specifically asking about the identities of civilian, not military, targets.
15. Clarke (2005) and Ray (2005) underscore the necessity of including relevant alternative variables in the statistical model.
16. Studies that use this database include Acosta (2014, 2016) and Barceló and Labzina (2020).
17. An example is the attempted use of female suicide bombers against Iraqi schools, as in the case of Rania Ebrahim Mutleg in Baqouba (oumpushkina.wordpress.com/tag/iraq/).
18. I also account for a mixed category of security and civilian targets, recognizing that some terrorist events may produce victims from both targets (e.g., a hospital on a military base and a civilian contractor wearing body armor and driving an armored truck). Such events do not easily fit into a pure security target category. I create another dependent variable by combining the pure security target cases with mixed cases. [Supplementary Appendix 1](#) displays results, which are very similar to those of the baseline models in [Table 2](#), indicating that gender differences matter.
19. When I incorporate cubic polynomials of time to account for potentially nonlinear trends in Security Target, the results remain similar.
20. Security targets are not clustered in certain terrorist organizations since they are not organization-specific.
21. Thomas (2021) applies the same operationalization to female suicide bombers, and Soules (2022) uses it as his dependent, not independent, variable.
22. Unknown gender cases account for 2,691 (about 44 percent) out of 6,127. When these unknown gender cases are excluded from the sample, the main findings of this study do not change in a meaningful way (see [Supplementary Appendix 4](#))—female bombers are less likely to strike security targets and more likely to go after civilian targets.
23. The view that the small percentage of women poses no real security threat overlooks the increasing rate of women’s participation, especially with the recent activities of terrorist organizations such as ISIS and Boko Haram.
24. I assess multicollinearity among the predictors using variance inflation factors, R^2 , eigenvalues, and condition index. There is no significant multicollinearity to report.
25. An alternative explanation posits that terrorist organizations strategically deploy available suicide bombers, opting for female bombers because their attacks against civilian targets generate more media attention than those carried out by male bombers. To test this theoretical conjecture, I construct a multiplicative regression model of the relationship between gender and media attention (for more detailed explanations, see [Supplementary Appendix 5](#) and [Supplementary Appendix Figure 1](#)).
26. See <https://www.efetokdemir.com/data.html>.
27. See <http://apps.start.umd.edu/gtd/>.
28. See <https://www.revolutionarymilitant.org/>.
29. Political command captures the degree of top leaders’ political command over the entirety of organizational operations, both military and non-military.
30. Thomas states that “given the difficulty of measuring gender norms directly, I employ three proxies. First, I include WOMEN’S CIVIL SOCIETY PARTICIPATION as a proxy for societal expectations about women’s involvement in political violence” (p. 781).
31. I also include GDP per capita as a control variable, as suggested by a reviewer.
32. Note that all model estimations consider the clustering effect.

33. The order condition for identification requires that the number of excluded exogenous variables (the additional instruments) be at least as high as the number of included endogenous variables. The Stata command, `ivprobit`, that I employ checks this identification problem and generates no error message, indicating the order condition is met.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

Data Availability Statement

The replication materials will be posted at <https://dataverse.harvard.edu>.

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